SIREN ALERT FEATURE - Audio Warning When a Match is Found

When the system detects a registered (known) person, besides sending the email and saving a snapshot, it now also plays a POLICE-STYLE SIREN sound as an audio alarm. This grabs immediate attention at the control room.

1. New File: src/siren.py

Class: SirenPlayer

- __init__(): stores the sound file path and creates it if missing
- _generate_siren_wav(): produces the siren sound WAV file using numpy + wave
- play(): plays the sound using the platform audio tool (aplay/paplay on Linux, winsound on Windows, afplay on Mac)
- play_async(): plays the sound in a BACKGROUND thread so the camera loop is not blocked
- test(): quickly plays the siren to verify your speakers work

How the Siren Sound is Created (no internet needed)

We generate a "wee-oo-wee-oo" police siren mathematically:

```
t = time samples (0..3 seconds)
freq = 500 + 450 * sin(2*pi*1.5*t)
       -> the pitch smoothly rises and falls between 500-950 Hz
wave = sin(2*pi * cumsum(freq))   -> build the sine wave
envelope -> fade in/out so there is no click at the edges
save as 44.1 kHz 16-bit stereo-compatible WAV file
```

2. How It Is Wired In

settings.py (new lines)

```
SIREN_ENABLED = True
SIREN_SOUND_PATH = sounds/siren.wav
```

surveillance.py (terminal version)

```
from src.siren import SirenPlayer
self.siren = SirenPlayer() if SIREN_ENABLED else None
...
when match detected:
    self.siren.play_async()   # one line added
```

dashboard.py (web version)

```
from src.siren import SirenPlayer
self.siren = SirenPlayer() if SIREN_ENABLED else None
...
when match detected:
    self.siren.play_async()   # one line added
```

Why play_async() ?

Playing a 3-second sound blocks the program for 3 seconds. If we call play() directly inside the camera loop, the live video would freeze for 3 seconds. play_async() starts a tiny background thread that plays the sound while the camera keeps running smoothly.

3. How to Run

```
# Test the siren alone:
/usr/bin/python3 -m src.siren

# Terminal surveillance (siren now rings on match):
/usr/bin/python3 -m src.surveillance

# Web dashboard (siren rings on match in the browser stream too):
/usr/bin/python3 app.py
```

4. To Turn the Siren OFF

Open config/settings.py and set SIREN_ENABLED = False

5. Files Touched

- NEW: src/siren.py - SirenPlayer class
- NEW: sounds/siren.wav - generated sound file (auto-created)
- EDIT: config/settings.py - SIREN_ENABLED, SIREN_SOUND_PATH
- EDIT: src/surveillance.py - play siren on match
- EDIT: src/dashboard.py - play siren on match